

## **Supplemental Material for “Robust recalibration of aggregate probability forecasts using meta-beliefs” by Cem Peker and Tom Wilkening.**

This document provides a guide to the supplemental material, which includes the data sets and R source files to replicate the results in the manuscript “Robust recalibration of aggregate probability forecasts using meta-beliefs” by Peker and Wilkening.

Results are generated using R version 4.3.1 and RStudio Version 2023.06.2+561. It is sufficient to run the commands in *recalibr\_analysis.R* in order to reproduce all figures and tables in the paper.

Unzip the supplemental material folder, open *recalibr\_analysis.R* and run the commands in order. The working directory should be the same directory as *recalibr\_analysis.R*. This is set by default if RStudio is launched by opening *recalibr\_analysis.R*. Inline comments in *recalibr\_analysis.R* provide more information on how the codes should be run and results produced by each command. Tables and statistical results are printed on the console. Figures are saved as pdf files in the figures folder.

### **recalibr\_analysis.R**

Includes the R scripts to reproduce all the tables, figures and statistical results in the manuscript. Inline comments in *recalibr\_analysis.R* provide information on how the codes should be run and the results produced by each command. Reads *recalibr\_dataset.csv* from the data folder.

### **recalibr\_dataset.csv**

Constructed combining the raw data from Wilkening et al. (2022) and Howe et al. (2024).

Variables:

- *task\_type*: “Science”, “States”, “Artwork” or “NFL”.
- *item*: Unique identifier for the task
- *subject*: Subject ID (not unique across task types)
- *prob*: Probability prediction, between 0 and 1
- *metaprob*: Metaprediction on average prob in that task, between 0 and 1
- *binary\_outcome*: Correct answer (binary), 0: False/No, 1: True/Yes

### **construct\_dataset.R**

Constructs *recalibr\_dataset.csv* using the raw data from Wilkening et al. (2022) and Howe et al. (2024). Reads raw data from *data/rawdata* folder. The output (*recalibr\_dataset.csv*) is saved in *data/recalibr\_dataset.csv*. This script does not have to be run for reproducing the results, the output (*recalibr\_dataset.csv*) is already included in the Data folder. Running this script overwrites *data/recalibr\_dataset.csv*.

### **Data folder**

Includes the data set *recalibr\_dataset.csv* constructed from raw data and two subfolders which include raw data. More information on the raw data below.

### **Rawdata subfolder in Data**

Includes two subfolders “wilkening2022” and “howe2024”, which provide the raw data.

Files in wilkening2022:

- *r2a\_rawdata.csv*, *r2b\_rawdata.csv*, *r2c\_rawdata.csv*, *r2d\_rawdata.csv*, *r2e\_rawdata.csv*: Responses to “Science” tasks, collected over 5 samples.
- *r2\_outcomes.csv*: Questions and binary truths for the Science tasks
- *m4data.csv*: Responses to “States” tasks
- *m4data\_details.csv*: Questions and binary truths for the States tasks
- See Wilkening et al. (2022) for more information.

Files in howe2024:

- *mwh\_data.csv*: Data from Howe et al. (2024). *Pr\_forecast* and *Meta-Prediction* provide the prediction and meta-prediction data.
- *mwh\_QuestionsList.csv*: Questions and binary truths for the tasks in the Howe et al. (2024) data.
- Subfolder *artwork*: Drawings that participants predict the market value of in the Artwork tasks (Howe et al., 2024).
- See Howe et al. (2024) for more information.

### **Figures folder**

Provides the figures (in png or pdf format) included in the accepted manuscript. Figures in Section 5 and Appendices D-F can be reproduced by running *recalibr\_analysis.R* (overwrites existing pdf files). See howe2024 subfolder in raw data for the images in Figure C1 in Appendix C. Other figures are made by using a drawing tool (<https://ipe.otfried.org/>).

## **References**

Howe, P. D., Martinie, M., & Wilkening, T. (2024). Using cross-domain expertise to aggregate forecasts when within-domain expertise is unknown. *Decision*, 11(1), 35-39. <https://doi.org/10.1037/dec0000212>

Wilkening, T., Martinie, M., & Howe, P. D. (2022). Hidden experts in the crowd: Using meta-predictions to leverage expertise in single-question prediction problems. *Management Science*, 68(1), 487-508. <https://doi.org/10.1287/mnsc.2020.3919>